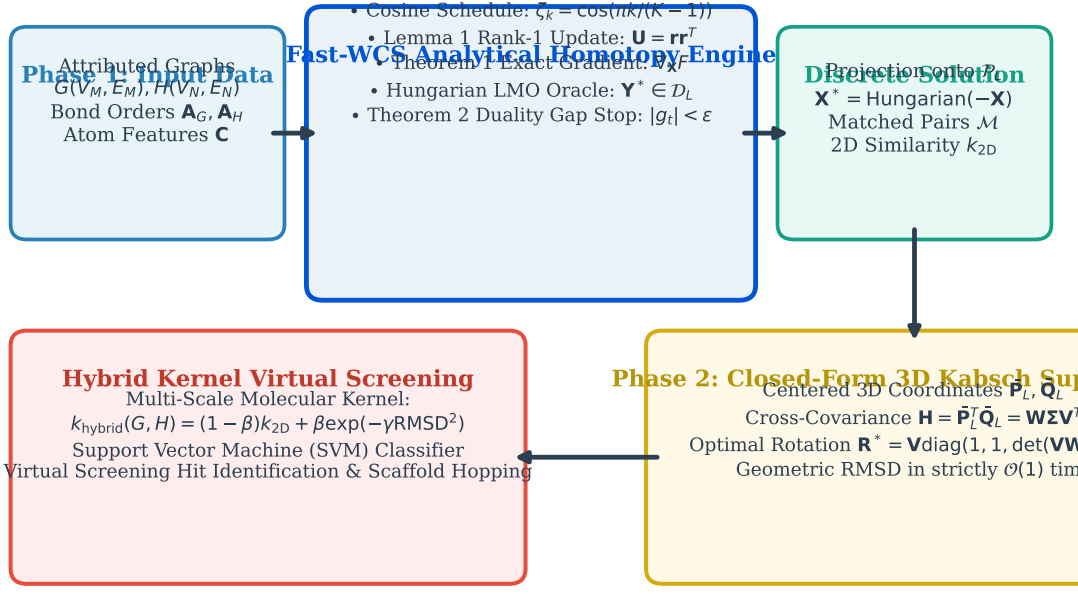

Phase 1: Input Data
Attributed Graphs
$G(V_M, E_M), H(V_N, E_N)$
Bond Orders $\mathbf{A}_G, \mathbf{A}_H$
Atom Features $\mathbf{C}$
Fast-WCS Analytical Homotopy Engine
• Cosine Schedule: $\zeta_k = \cos(\pi k/(K-1))$
• Lemma 1 Rank-1 Update: $\mathbf{U} = \mathbf{r}\mathbf{r}^T$
• Theorem 1 Exact Gradient: $\nabla_{\mathbf{X}} F$
• Hungarian LMO Oracle: $\mathbf{Y}^* \in \mathcal{D}_L$
• Theorem 2 Duality Gap Stop: $|g_t| < \varepsilon$
Discrete Solution
Projection Center $\mathcal{P}$
$\mathbf{X}^* = \text{Hungarian}(-\mathbf{X})$
Matched Pairs $\mathcal{M}$
2D Similarity k_{2D}
Phase 2: Closed-Form 3D Kabsch Superposition
Centered 3D Coordinates $\bar{\mathbf{P}}_L, \bar{\mathbf{Q}}_L$
Cross-Covariance $\mathbf{H} = \bar{\mathbf{P}}_L^T \bar{\mathbf{Q}}_L = \mathbf{W}\mathbf{\Sigma}\mathbf{V}^T$
Optimal Rotation $\mathbf{R}^* = \mathbf{V}\text{diag}(1, 1, \det(\mathbf{V}\mathbf{W}^T))\mathbf{W}^T$
Geometric RMSD in strictly $\mathcal{O}(1)$ time
Hybrid Kernel Virtual Screening
Multi-Scale Molecular Kernel:
$k_{\text{hybrid}}(G, H) = (1-\beta)k_{\text{2D}} + \beta\exp(-\gamma\text{RMSD}^2)$
Support Vector Machine (SVM) Classifier
Virtual Screening Hit Identification & Scaffold Hopping